

Requirements Table — Pharmacy Expiry & Re-order Dispatch Engine

PESIUG24CS125 | Charan M | Section B | Problem Statement #15 — Healthcare & Telemedicine

FR-001	Functional	The system shall enforce FEFO (First-Expired-First-Out) picking order by prioritizing medicine batches with the nearest expiry date during sales checkout.	High	Pass: Oldest valid (non-expired) batch is selected first for dispensing. Fail: A newer batch is assigned while an older, unexpired batch of the same medicine still exists in stock.	Minimizes medicine wastage from expiry and ensures patient safety by moving older stock first (given requirement).
FR-002	Functional	The system shall require a Pharmacy Clerk to record a medicine batch's batch number, quantity received, and expiry date at the time of stock receipt.	High	Pass: A newly received batch cannot be saved to inventory unless batch number, quantity, and a valid future expiry date are all entered. Fail: A batch is saved to inventory with any of these fields blank or with an expiry date in the past.	Accurate batch-level expiry data is the foundation the FEFO engine and expiry alerts depend on.
FR-003	Functional	The system shall generate a daily expiry alert list showing all batches expiring within the next 30 days, grouped by medicine.	Medium	Pass: Running the alert job on a given day returns every batch whose expiry date falls within 30 days of that day, with none omitted. Fail: The list is not generated on schedule, or a batch expiring within 30 days is missing from it.	Gives pharmacy staff advance warning to prioritize dispensing or return/markdown of near-expiry stock.
FR-004	Functional	The system shall allow a Pharmacy Clerk to manually override the FEFO-suggested batch at checkout, provided a justification note is entered.	Medium	Pass: An override is accepted only when a non-empty justification note is supplied, and the override is logged with clerk ID, timestamp, and reason. Fail: An override is accepted without a justification note being recorded.	Handles legitimate exceptions (e.g., damaged packaging on the oldest batch) while preserving an audit trail.
FR-005	Functional	The system shall allow an Inventory Supplier to acknowledge a dispatched purchase order and provide an expected delivery date.	Medium	Pass: A purchase order's status changes to "Acknowledged" with an expected delivery date once the supplier confirms. Fail: The order remains in "Sent" status indefinitely with no way to record supplier acknowledgement.	Closes the loop on the re-order process so pharmacy staff can track incoming stock.
NFR-001	Nonfunctional (Performance & Security)	The system shall automatically generate and dispatch supplier purchase order requests when item stock dips below the safety threshold.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load. Fail: Purchase order dispatch exceeds the target latency or fails a security benchmark under peak load.	Prevents stock-outs of essential medicines while keeping the automated dispatch channel secure (given requirement).
NFR-002	Nonfunctional (Compliance & Auditability)	The system shall retain a complete, tamper-evident audit trail of batch expiry data and dispensing transactions for a minimum of 3 years.	High	Pass: A dispense/expiry record created today remains retrievable and unaltered when queried 3 years later in an audit test. Fail: A record is purged before 3 years, or can be edited without the change being logged.	Required for pharmaceutical regulatory compliance and traceability of dispensed medicine batches.

GITHUB LINK: [charanmahesh/SE-Labs-PES1UG24CS125](https://github.com/charanmahesh/SE-Labs-PES1UG24CS125)